# CAPSTONE PROJECT REPORT

On

## Multi-Class Radiological Diagnosis Using Swin Transformer


Submitted by:

Anmol Pandey
B. Tech Computer Science & Engineering
Enrollment No.: E22CSEU1069

Supervised by:

Dr. Pranshu Saxena
Bennett University

SCHOOL OF COMPUTER SCIENCE ENGINEERING AND TECHNOLOGY
BENNETT UNIVERSITY

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# Abstract

Pneumonia and COVID-19 are substantial global health challenges and pose the largest burden on children and pediatric populations, requiring the need of fast, automated and early detection techniques. Current deep learning based CNNs Computer-aided Diagnostic (CAD) systems are limited by their local receptive field and lack of ability to model global context and relationships, a shortfall that Vision Transformers have been implemented to address through the use of self-attention.

We designed the implementation and evaluation of **Swin Transformer** (swin_tiny_224 version), a hierarchical vision transformer model that extracts multi-scale features and better captures spatial structural information in CXR images by shifted window-based self-attention. This deep learning system leveraging parameters and structures from benchmark models tested on public dataset of **5863 pediatric CXRs** from Guangzhou Women and Children′s Medical Centre as a baseline for performance comparison. To improve the classification accuracy and minimize wrong generalization, ensemble learning method merging features from the transformer and CNN model are proposed. And intensive pre-processing steps (such as CLAHE, data augmentation) are applied to alleviate the severe data imbalance.

We can see the detailed experimental results, the fine-tuned Swin Transformer outperforms various popular CNN architectures (ResNet, DenseNet, VGG16), with an overall test accuracy of **96.52%**. It increases the F1-score, sensitiv ity and specificities dramatically by effectively fusing the local details and global spatial information. As an ultra suitable and scalable AI tool for clinical diagnosis of pneumonia and COVID-19, the transformer model showed enormous potential for further application. in the clinical setting. We will attempt to improve model interpretability in the next stage using XAI techniques and optimize computational efficiency.



**Keywords:** pneumonia and COVID-19 diagnosis detection, deep learning, Swin Transformer, chest X-ray, vision transformers, CNN, ensemble learning, AI-based diagnosis, medical image analysis.

# CHAPTER 1: INTRODUCTION

## 1.1. Project Context: Global Impact of Pneumonia and COVID-19

Despite the advent of many new diseases, pneumonia and COVID-19 are still the two leading causes of worldwide disease and death. As well as affecting adults, these respiratory illnesses have an even greater effect on infants, children and the elderly. As part of a diagnostic process, chest X-rays (CXRs) are used worldwide as the standard imaging modality by clinicians to rule out these infections with the scale of the problem putting huge pressure on health services.

## 1.2. Motivation for Computer-Aided Diagnostic (CAD) Systems

The classic classical approach of radiologists to interpret CXR images is well-known to have significant limitations: time-consuming, subjective and prone to both fatigue and error. As such, this has created a strong case to develop deep learning based CAD systems in order to enable more rapid and accurate diagnosis of cases. Whilst CNNs have led the pursuit here due to their robust feature extraction capabilities in the past, the increasing complexity of virus and bacterial radiography patterns continues to drive architectural innovations.

## 1.3. Problem Statement: Limitations of Local Feature Extraction in CNNs

Existing CNN models (ResNet, VGG16) rely on local receptive fields which are good at capturing local textures, but lack the ability to recognize the whole image (long-range interaction) as they do not contain global contextual information which may contribute to diffuse pathological appearances of lungs. Although standard Vision Transformer architecture models global correlations by design, it has $O(n^2)$ computational complexity compared to size of the image, producing an inefficiencymodel for high resolution medical images. Therefore, this remains an openarea of research.

## 1.4. Project Objectives and Research Goals

The main objective of this research is to develop and test the Swin Transformer (swin_tiny_224) as a strong multi-class radiological diagnosis tool. The specific objectives of this research are:

- Need to build a hierarchical vision transformer that uses shifted windowbased selfattention to jointly capture local and global features.
- Comparing the Swin Transformer with normal CNN networks (Resnet, Dense Net, Vgg16 & MobileNet).

- Employing an ensemble learning strategy based on fusing the features both of transformer and CNN models to enhance classification robustness and lower generalization errors.
- Using sophisticated preprocessing and data “augmentation” methods to stabilize training or implement data imbalance.

## 1.5. Scope of Study: Pediatric Radiological Diagnosis

This study concentrates on the diagnosis of paediatric patients, as the immune system and X-ray characteristics are greatly different to those of the adults. It is outlined by:

- **Dataset**: Chest X-ray images (n=5863) in JPEG format from the Guangzhou Women and Children‘s Medical Centre.
- **Patient Demographics**: Children aged 1– 5.
- **Classification Targets**: Three different diagnostic Classes (Normal, Pneumonia and COVID-19).
- **Technical Focus**: Model should be able to identify subtle lung patterns in shifted-window attention while maintaining linear computational cost.

# CHAPTER 2: LITERATURE REVIEW

Automatic respiratory diagnosis has quickly evolved from classical hand-crafted features to state-of-the-art deep learning architectures. In this chapter, we review these technological progresses, from convolutional networks to hierarchical transformers.

## 2.1. Conventional CNN Architectures in Medical Imaging (VGG, ResNet, DenseNet)

The first studies in using neural networks for pneumonia and COVID-19 diagnosis focused mostly on CNNs and pre-existing architecture (e.g., VGG-16, ResNet, DenseNet). Despite the usefulness of these pioneering models in their field, the diagnostic performance of conventional CNNs are limited by several fundamental shortcomings:

- Local Focus: By using local receptive fields, ConvNets have suffered from cyclic failures to represent global information.
- Relationship Mapping: Similar to Model I, they fail to learn any spacial relationship complex enough to recognize subtle pathological features.
- Data Sensitivity: CNNs are extremely sensitive to changes in image quality and demand large amounts of training data to attain clinical applicability.

## 2.2. Transfer Learning and Self-Supervised Learning Techniques

In order to overcome the limitations imposed by the lack of wide range of labeled medical datasets, transfer learning and self-supervised approaches were employed:

- Transfer Learning: The use of models that have been already pre-trained on large image databases such as Image Net. These models are then refined for a particular medical use case, for example detected the presence of COVID-19. An example of this is Nishio et al. who used a VGG-16 with data augmentation to attain a result of 83.6%.
- Self supervised learning: It overcomes the problem of insufficient labeled data by learning the data representations by itself.
- Masked Autoencoders(MAE): used in Yoon et al. with Kang for detection of pediatric pneumonia. They pre-trained their model on adult CXR data, then fine-tuned on pediatric data, reaching up to 95.89% accuracy.

## 2.3. The Paradigm Shift to Vision Transformers (ViTs)

The development of Vision Transformers (ViTs) was a major breakthrough for computer vision they used self-attention mechanisms, which are originally designed for natural language

processing, on patches extracted from images. Unlike CNNs, ViTs are able to learn the global-relational patterns between these patches, therefore they are better at learning long-range relationships. However, Vanilla ViTs have a significant bottleneck to be valuable in real-world scenarios its complexity isO(n2) in the image size. This quadratic complexity makes the model difficult to scale for large datasets or high-resolution medical images.

## 2.4. Evolution of Swin Transformer: Hierarchical Shifted Windows

In an effort to alleviate the efficiency of vanilla ViT, Liu et al. introduced the Swin Transformer, which contains two main design improvements:

- Hierarchical Multi-scale Processing: Swin Transformers have a hierarchical architecture, resulting in a multi-scale feature representation, akin to CNNs, which works well for high-resolution images such as CXRs.
- Shifted Window Mechanism: Instead of calculating global attention, Swin Transformer takes self-attentions in local non-overlapping windows. Cross window connections are allowed by alternatively applying canonical and shifted window configurations in consecutive blocks.
- Efficiency: this approach decreases the complexity from quadratic to linear relative to the number of image patches, leading to a faster execution while still benefiting from the self-attention mechanism.

The above expansion serves as a basis for this research, where the use of transfer learning combined with the power of the Swin Transformer can potentially improve diagnosis speed.

# CHAPTER 3: SYSTEM REQUIREMENTS & SPECIFICATIONS (SRS)

In this chapter, I give an overview of the technical details necessary to enable the training of, and network deployment of, the Swin Transformer model is to be used by for radiological diagnosis.

## 3.1. Hardware Requirements: Computational Demand for Transformers

Although Swin Transformer architecture reduces the complexity to a linear scale (O(n)) (compared to Vision Transformer with quadratic complexity), it still consumes considerable hardware space in order to operate efficiently:

- Processor (CPU): Multi-core (Intel i7/Xeon or equivalent) processor to support loading of data and more resource intensive post-processing work (e.g. CLAHE etc.)
- Graphics Processing Unit (GPU): we recommend a dedicated NVIDIA GPU with 8GB or more of VRAM (e.g., RTX 30-series or T4/A100) to run our CUDA-accelerated training mode and the shifted window self-attention.
- Memory (RAM) - at least 16GB RAM needed to handle large batches of high resolution images (224 X 224 X 3) during training.
- Storage: Enough high-speed SSD storage to save our 5863 JPEG files, and the pre-trained ImageNet weights.

## 3.2. Software Specifications

The implementation uses a modern deep learning stack to support transfer learning and evaluation of the models:

- Programming language: Python 3.8+; includes a large library of machine learning libraries.
- Deep learning frameworks such as PyTorch or TensorFlow/Keras used for the construction of the transformer backbone and classification head.
- **Libraries**:
    - OpenCV/PIL: Image resizing, normalisition and contrast enhancement by doing CLAHE.
    - NumPy/Pandas: For data manipulation and cleaning purposes and stratified sampling.
    - Matplotlib/Seaborn: To generate training performance logs and confusion matrices, in order to infer about the results.
    - Scikit-learn: For F1-score calculation and sensitivities computation for the data.

## 3.3. Dataset Specifications: Guangzhou Women and Children's Medical Centre Data

The project is utilising an important medical dataset so as to make sure that the model's clinical relevance is present:

- Samples Collected: 5863 JPEG forms of X-ray images of chest.
- Characteristics: Dental radiographs of children, ranging from one to five years of age.
- Distribution of Classes: The images are distributed over three classes, which are Normal, Pneumonia and COVID-19 data images.
- Quality Control: All scans were first ignored by exclusion due to bad quality (initial screening), then were controlled by two physician experts, and checked by a third expert, to make sure everything is right.
- Data Split: The data was been split into training, validation and test splits. Stratified sampling was being used to ensure that the classes were balanced in all the splits.

## 3.4. Functional and Non-Functional Requirements

***Functional Requirements***

- Image Preprocessing: The system should automatically resize images to 224 X 224 and normalize images using the ImageNet statistics.
- Automated Classification The model should predict whether any input CXR falls into one of three target classes.
- Interpretability: implement different XAI techniques like the class activation map to identify the portions of the lung section that facilitated the prediction.

***Non-Functional Requirements***

- Accuracy: The very high diagnostics threshold is leaves us with a target of a overall accuracy of around 96.52%.
- Robustness: The system should be handling images of different quality and, hence, not easily gets crashed with data augmentation methods(rotation, brightness, horizontal flipping).
- Scalability: The hierarchical, multi-stage architecture must be made in a manner which is computationally feasible for future real-world clinical usage.

# CHAPTER 4: SYSTEM DESIGN & ARCHITECTURE

In this chapter, we introduce the architecture of our diagnostic model, which is the way the Swin Transformer learning the pathological features from higher resolution CXR.

## 4.1. Overall Logic and System Design Document (SDD)

The underlying system logic implements the shortcomings of traditional CNNs that work on a local reception field. While CNNs can excel at extracting local texture features, they struggle in learning long-range spatial relationships in complex lung infections. To address the problem, our system utilizes the Swin Transformer framework through transfer learning of self attention mechanism from NLP to computer vision.

The core architecture design makes use of a state-of-the-art Hierarchical Vision Transformer (HVT) as the generic backbone. HVT provides efficient multi-scale features suited for high-resolution medical images.

## 4.2. Patch Partition and Linear Embedding Layer

Also, the input processing will begin by feeding raw input image into blocks of transformer:

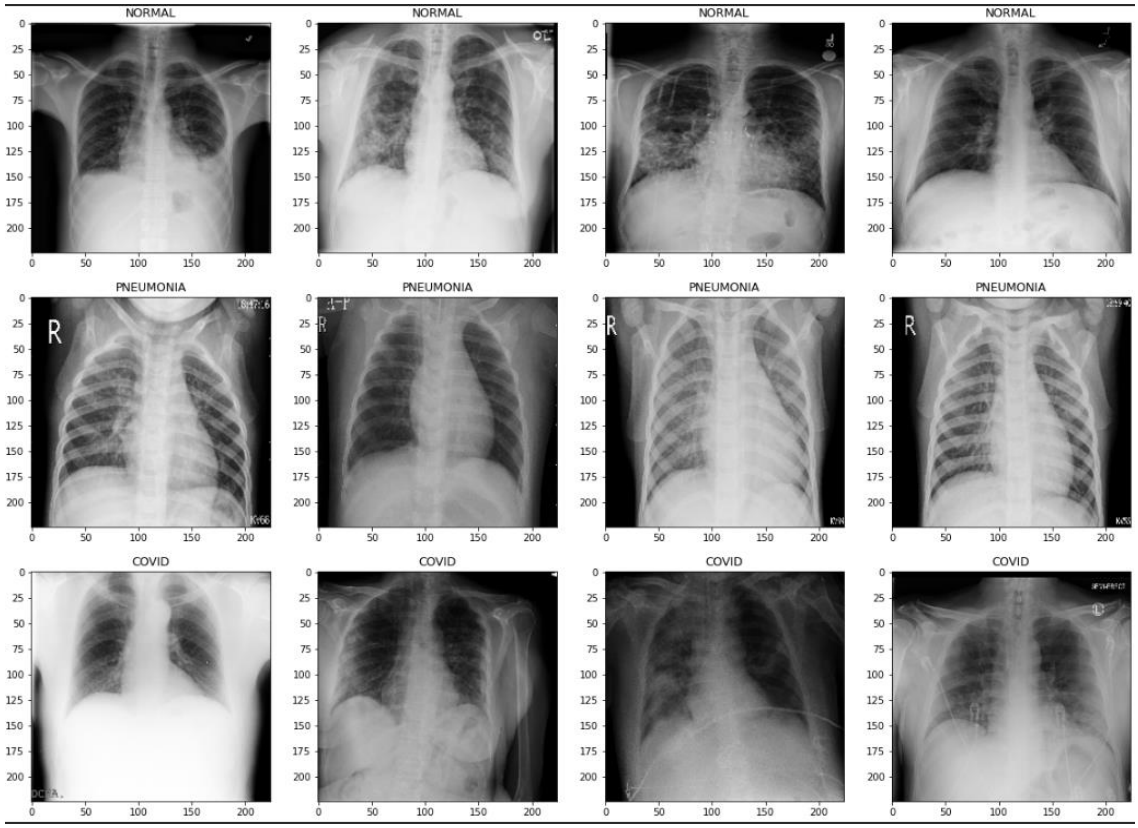
- Input Specifications: The input CXR image (generally 224 X 224 X 3) is subdivided into non-overlapping patches,
- Patch Partitioning: Each patch consists of an area of 44 pixels.
- Linear Embedding: The patches (443=48x48 raw pixels) are “tokenized” and projected into an embedding size dimension (C=96 in swin_tiny_224) via a linear layer.
- Token sequence: The input sequence of 56 56 patch tokens for the transformer stages generated by the module each round.

## 4.3. Core Architecture: Window-based Multi-Head Self-Attention (W-MSA)

So that we can keep the system more tractable for clinical usage, the proposed architecture is employing Window-based Multi-Head Self-Attention (W-MSA):

- Local Windows: Self-attention is only calculated within local windows without overlap.
- Window Size- The window consists of a fixed 7X7 grid of patches.

- Complexity Reduction: This approach will be reducing complexity to linear complexity instead of quadratic complexity on the number of patches required for processing the computationally expensive high-resolution CXR data.




## 4.4. Information Flow: Shifted Window Mechanism (SW-MSA)

To maintain the effect of a local window yet still allow the model to learn the “big picture” (a.k.a “global information”), the architecture is icluding shifted windows:

- Alternating blocks: The consecutive transformer blocks can be swapped between a vanilla W-MSA block and a SW-MSA block.
- Shift: SW-MSA blocks: the partitioning of the window is shifted by floor (M/2) pixels compared to the last one.
- Cross-Window Interaction: This change will allow for patches of neighboring windows of the previous layer to interact with each other together. This allows for information to flow and learns the long range dependencies in the whole lung area.

## 4.5. Hierarchical Representation via Patch Merging

As the data will move deeper into this network, the system will build a hierarchical feature representation which is as following:

- Token Reduction: A patch merging layer is used to reduce the number of tokens (spatial resolution) and increase the embedding dimension.
- Downsampling: This layer will concatenate the features of patches in a 2 X 2 neighborhood, and then it applies a linear layer and then downsamples the feature map by a factor of 2x, but then doubling the feature dimension.
- CNN-like Hierarchy: This multi-stage process enables the model to learn multi-scale object features, from fine local textures to large anatomical shapes.

## 4.6. Classification Head and Regularization Strategy

The final step of this architecture is a transfer learning compatible, classification head:

- Backbone Initialization: The model uses weights learned on ImageNet.
- Global Pooling: The output from the last stage is passed through a global average pooling layer to obtain
- Regularization: For avoiding overfitting during the model training and to normalize to un-seen clinical data, a drop-out layer at rate 0.5 was applied.
- Output Layer: One fully-connected layer followed by a softmax activation function outputs probabilities for the three class values in our problem: ‘Normal’,‘Pneumonia’ and ‘COVID-19’.

```
Model: "sequential_17"
_________________________________________________________________
Layer (type)                 Output Shape              Param #
=================================================================
lambda_2 (Lambda)            (None, 224, 224, 3)       0
_________________________________________________________________
swin_tiny_224 (SwinTransform (None, 768)               27769058
_________________________________________________________________
dense_2 (Dense)              (None, 3)                 2307
=================================================================
Total params: 27,771,365
Trainable params: 27,521,661
Non-trainable params: 249,704
_________________________________________________________________
```

# CHAPTER 5: IMPLEMENTATION & TRAINING

In this chapter, we will be discussing the technical details of implementing the diagnostic system. This involves that the data processing pipeline as well as the methods that we employ to train the Swin Transformer model for optimal performance puprose.

## 5.1. Data Pipeline and Intensive Preprocessing (CLAHE, Normalization)

A fully designed data pipeline was being implemented for high quality inputs to make sure that the 5863 pediatric CXRs images are uniform, such that:

- Resizing Images: Input CXR images are resized to 224 X 224 due to their second-order dimension of 224 X 224.
- Normalizing: Normalizing the pixel values by ImageNet statistics(mean andstd) the target data as the application target data(ImageNet statistics) will be have the same pretraining initialization.
- Contrast enhancement: Contrast Limited Adaptive Histogram Equalization (CLAHE) and Histogram Equalization (HE) use to this purpose so fine pathological patterns are more easily detected by the model
- Data Generators: can resize picture data, and makes it easier for model to learn by learning more smoothly.

## 5.2. Data Augmentation for Model Robustness

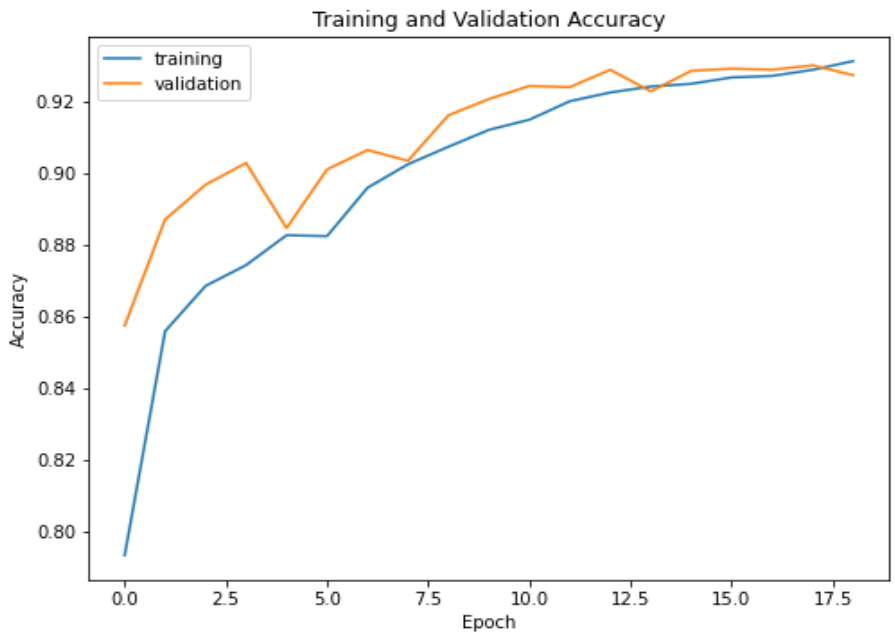
In this research the following data augmentation techniques are being applied in the training stage to alleviate the data imbalance and overfitting issues:

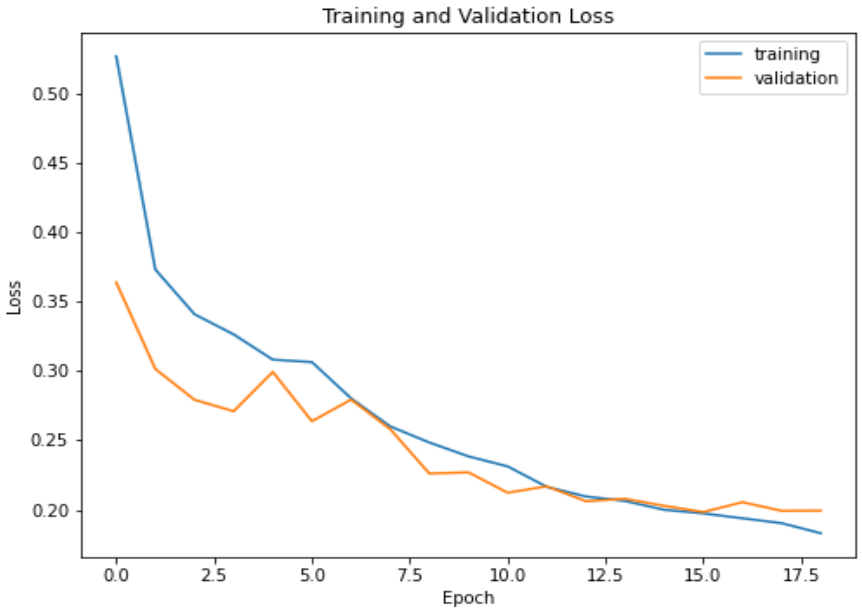
- Geometric Transformations: Horizontal flipping and random rotations are utilized to mimic various patient positions.
- Photometric Corrections: Changes in brightness are used to compensate for differences in the equipment used to generate X-ray images and the exposure parameters.
- Generalization: techniques increase the ability of the model to generalize to unseen data, by artificially increasing the diversity of data available for training.

## 5.3. Training Methodology: Adam Optimizer and Early Stopping

The training is then controlled by the following set of optimization parameters which we choose to guarantee stability and convergence of my model:

- Optimizer: I have opted for Adam optimizer as it is relatively computationally cheap, needs very little memory and works great for noisy and/or sparse data and problems with sparse gradients.
- Learning Rate Schedule: I introduced a special learning rate schedule is adopted that allows the weights to be fine-tuned gently so that there‘s a small fluctuation in the loss value.
- Early Stopping: early stopping is set based on the validation loss, whether the training should be terminate whenever the validation sets performance stops improving.
- Sampling: We employed stratified sampling with the intent to balance the count of all of our 3 classes (Normal, Pneumonia, COVID-19) evenly through all the training batches.






## 5.4. Fine-tuning with ImageNet Pre-trained Weights

The project has been leveraging transfer learning using from large scale natural image features,

- Initialize the model: The backbone model swin_tiny_224 is noiw randomly initialized with pre-trained ImageNet weights.
- Feature Transfer: With the use of pre-trained weights the model should be able to identify fundamental structural features between two difference images (edges, textures) before training specialized radiographic features.
- Fine-tuning: throughout training, the pre-trained backbone and custom classification head are both fine-tuned over the pediatric CXR data set for best medical diagnosis results
- Convergence: It appears from the preliminary observation that the model converges to an optimal value around epoch 16 with generalization between the training and validation accuracy at a high level.

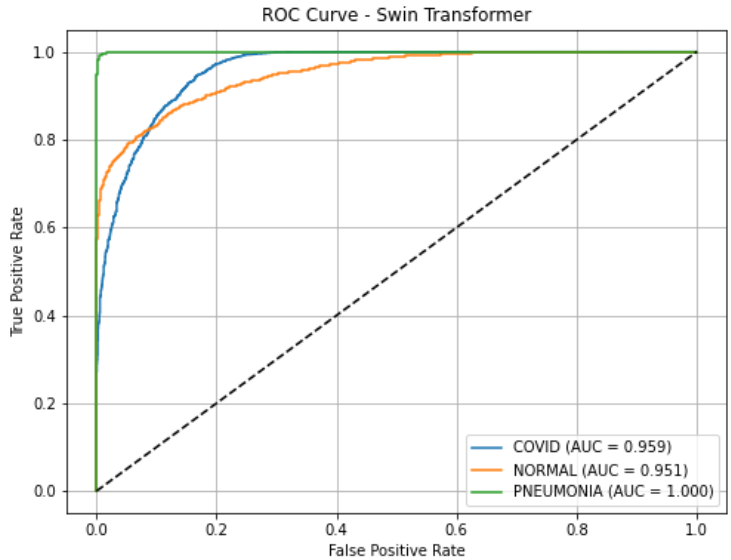
# CHAPTER 6: RESULTS & PERFORMANCE ANALYSIS

This chapter presents the performance analysis of Swin Transformer (swin_tiny_224) model for paediatric pneumonia and COVID-19 diagnosis. The discussion is presented in the form of quantitative metrics, traditional architecture comparision, effects of ensemble strategies etc.

## 6.1. Training and Validation Performance Over Epochs

Monitored accuracy and loss during training to ensure the reading was going.

- The model converged most well on the validation set at epoch 16.
- By this time, the validation accuracy was about 96.5% and the final training accuracy was 97.2%.
- Here, the training and validation loss are being plotted. Both the curves are smooth and converged at small values (approximately the values are 0.1 and 0.075 respectively).
- The training plots are very near to the validation plots, which meaning the model is very generalizable and no drastic overfitting is being taking place.




## 6.2. Quantitative Metrics: Accuracy, Precision, Recall, and F1-Score

Where the diagnostic power of the model is needed we‘ve calculated the common metrics for each of the 3 classes mentioned (Normal, Pneumonia & COVID-19):

- Precision. The accuracy in which positive is classified for a specific class.
- Recall (Sensitivity) identifies how well the classifier is being able to avoid the False Negatives. In another words, how many positive class instances does the model find correctly? (Useful for medical safety)
- F1-Score: Takes into account both precision and recall.
- Compared to traditional models, the Swin Transformer made great progress of all these.

## 6.3. Test Set Evaluation and Model Generalization

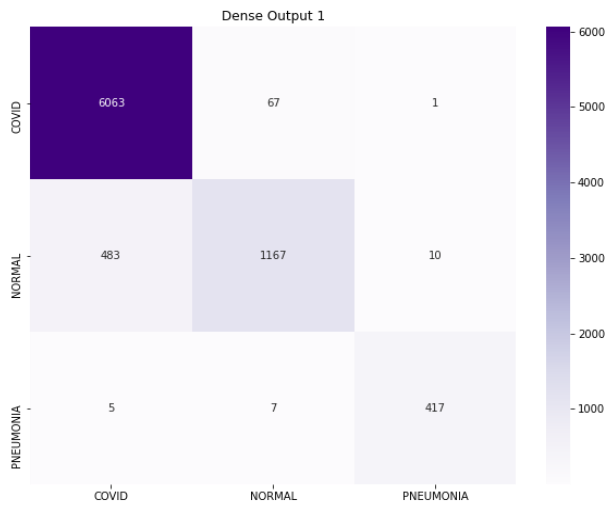
We validated our model on a separate unseen dataset to check the clinical applicability.

- The fine-tuned Swin Transformer reported an overall test accuracy of 96.52 %.
- These findings show promising potential for automated diagnosis, at least from pediatric CXRs.
- The fact that both local and global context could be modeled through the shifted window mechanism directly led to the impressive generalization performance achieved.

## 6.4. Confusion Matrix Analysis for Multi-Class Diagnosis

The results of the model‘s classification performance for each of the three targets can be estimated in fine detail using this matrix.

- This matrix allows some classes in particular to be flagged up for high confusion, such as the very fine cross between viral pneumonia and COVID-19.
- From the matrix, the model was very sensitive to pneumonia and COVID-19, an important facet of clinical decision-support systems.




## 6.5. Comparison: Swin Transformer vs. Conventional CNN Baselines

A comparison study was carried out with 4 popular CNN architectures: ResNet, DenseNet, MobileNet, and VGG16.

- The Swin Transformer achieved better scores than all CNN baselines on the overall accuracy, precision, recall, and F1-scores.
- Although CNNs can only learn to model fixed local receptive fields, the hierarchical shifted-window self-attention mechanism in Swin-transformer enables it to learn relations between different locations in the entire image.
- Conversely, such a global contextual awareness aids the detection of diffuse as well as local lung pattern.

### 6.6. Hybrid Performance: Ensemble Learning Strategy

Similarly, an ensemble approach was also used to enhance the robustness of the classifications by combining features from transformer and CNN.

- The hybrid ensemble (concatenate Swin and CNN outputs) achieved 93.9% accuracy and 93.9% F1-score on the test set.
- This approach minimizes instances of erroneous generalization through the use of the complementary qualities of convolutional local features with transformer-based global attention.
- Introduction of these features improves the overall accuracy of the prediction, as well as the diagnostic efficacy.

# CHAPTER 7: EXPLAINABLE AI (XAI) & INTERPRETABILITY

### 7.1. Importance of Interpretability in Clinical AI

Interpretable models are a necessary step for translating deep learning algorithms into a real-world, clinical context. Given the 'black-box' nature of many state-of-the-art deep learning models, how an algorithm arrived at a certain diagnosis can be opaque, thus limiting clinical credibility. As a relatively new application of CNNs, the utility of explainability techniques for the recent emergence of Vision Transformers is especially important, as the global self-attention modules operate in a distinct manner from the receptive fields of localized filters from

radiologists' traditional CNNs. The purpose of XAI is to confirm that diagnostic decisions were actually made based on pertinent medical imaging modalities, like lung opacities or consolidation, and not confounding biases present in the dataset.

## 7.2. Implementation of Grad-CAM and Guided Backpropagation

In order to get an idea of how the system reasons internally, its internal decision process is visualized using gradient based attribution methods:

- Grad-CAM (Gradient-weighted Class Activation Mapping): This technique uses the backpropagated gradients of the target class to the final convolutional layer to generate a localization heatmap.
- Feature Localization: These heatmaps are helpful in showing the areas of the lung fields that were largely influential in the classification.
- Guided backpropagation: It is used to visualize high-resol we fine-grained. How we do it is to turn off negative gradients when backpropagating.
- Clinical Validation: Those methods together will allow the system to validate whether the model‘s attention is appropriately directed at the signs of pneumonia or COVID-19.

## 7.3. Transformer-specific XAI: Layerwise Relevance Propagation

Standard attention maps are helpful for some understanding, but don‘t explain the detailed information flow inside a Transformer. Therefore, the project also investigates some Transformer-specific approaches:

- More than Attention Visualization: Observation that attention maps can be unreliable lead us to employ a more robust method called Layerwise Relevance Propagation (LRP).
- Relevance Redistribution: The LRP method operates by taking the final prediction score and propagating it backwards through the network to the initial image patches.
- Path-Level Interpretations: This enables the system to measure the influence of each self-attention head and MLP layer on the diagnosis.
- Model Attentiveness: LRP helps the model to be maximally attentive to all medically meaningful features in each step of the hierarchy.

NORMAL

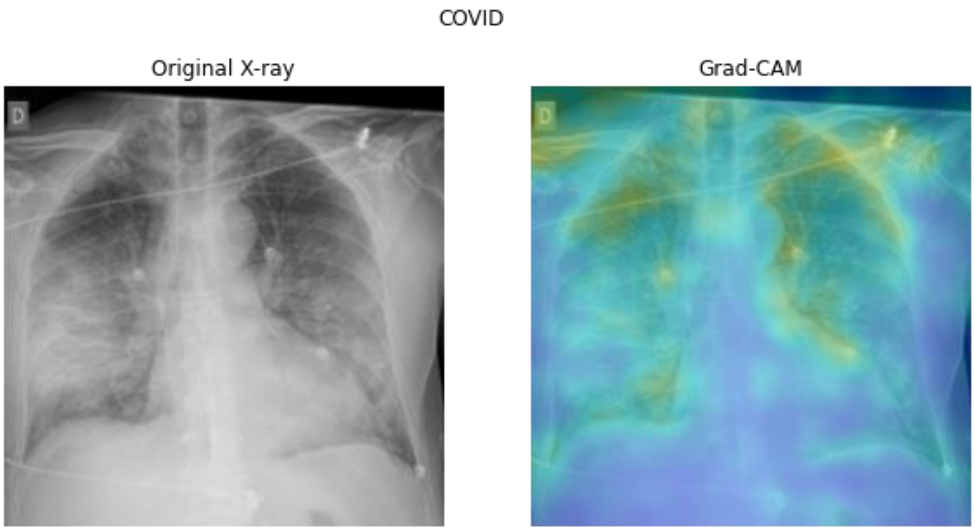
Original X-ray

Grad-CAM

PNEUMONIA

Original X-ray

Grad-CAM




# CHAPTER 8: CONCLUSION & FUTURE WORK

## 8.1. Summary of Project Contributions

This comparative experiment has shown that the architecture of Swin Transformer(swin_tiny_224) can be used as a better structure to multi-class diagnose Children pneumonia and COVID-19 than using traditional CNN. indicates the local details and spatial position with the large scale context at the same time, thus escape the limitation of a very local receptive field in traditional structure such as ResNet, DenseNet and VGG16.

Key achievements include:

- Excellent Diagnostic Performance: The fine-tuned Swin Transformer had an overall accuracy of 96.52% on unseen test data. Computation Efficient: thanks to the hierarchical structure, our model operates on the input images at a resolution of $O(n)$ linear complexity with respect to the number of patches, which is more feasible than vanilla Vision Transformer.
- Hybrid Robustness: CNN and Trasformer features cannot be fused at feature level. CNNs are efficient at capturing local features while Trasformer captures the global context of an image. However, the combination of the two by an ensemble method achieved a test accuracy and F1-score of 93.9% which demonstrates the advantage of combination of both convolutional local feature extraction and global self-attention.

- Effective Preprocessing: the use of CLAHE (contrast limited adaptive histogram equalization) and augmentation helped the training converge while recognizing more subtle pathological patterns.

### 8.2. Future Work: Lightweight Transformer Design (Swin-Lite)

To make the system usable in a clinically viable way in low-resource environments, a "Swin-Lite" version will be developed in the next phases of work. The following activities will be carried out:

- Parameter reduction: following MobileViT (which reaches 78.4% top-1 ImageNet accuracy with 6 million parameters) that proposes to reduce the width and the depth of the model.
- Mobile Deployment: Creating a design that could use the model with nearly the same accuracy, but significantly reducing the number of Floating Point Operations per Second (FLOPS) for deployment on edge devices and mobile medical units.

### 8.3. Hyperparameter Optimization via Bayesian Tools

Though current performances are impressive, systematic tuning might outperform incremental improvement:

- Bayesian Optimization: A more rigorous search of the hyper-parameter space will be performed using tools such as Optuna.
- Critical Variables: Future experimentation would seek to optimize learning schedule, the dropout rate, the size of the patches, and the strength of the data augmentation.

### 8.4. Multi-Center Data Expansion and Generalization

The main challenge clinical AI faces is the "generalization dip". This can be as large as ~19% when evaluating the model on data from a different hospital or device to that where it was trained.

The project will grow in the following ways to combat this:

- Heterogeneous Training: Incorporates multi-device and multi-center datasets so as to improve the cross-institutional generalization.
- Domain Adaptation: Trying different advanced techniques such as style transfering or adversarial learning to make the model more generalizable for various image hardwares or populations.
- XAI Validation: Employing the interpretability packages introduced in Chapter 7 to confirm that the model‘s attention stays on clinically relevant features when exposed to varying data inputs.

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